



Comparative efficacy and safety of first-line antiretroviral therapy for the treatment of HIV infection: a systematic review and network meta-analysis

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Summary

Background New antiretroviral therapy (ART) regimens for HIV could improve clinical outcomes for patients. To inform global guidelines, we aimed to assess the comparative effectiveness of recommended ART regimens for HIV in ART-naïve patients.

Methods For this systematic review and network meta-analysis, we searched for randomised clinical trials published up to July 5, 2015, comparing recommended antiretroviral regimens in treatment-naïve adults and adolescents (aged 12 years or older) with HIV. We extracted data on trial and patient characteristics, and the following primary outcomes: viral suppression, mortality, AIDS defining illnesses, discontinuations, discontinuations due to adverse events, and serious adverse events. We synthesised data using network meta-analyses in a Bayesian framework and included older treatments, such as indinavir, to serve as connecting nodes. We defined network nodes in terms of specific antivirals rather than specific ART regimens. We categorised backbone regimens and adjusted for them through group-specific meta-regression. We used the GRADE framework to interpret the strength of inference.

Findings We identified 5865 citations through database searches and other sources, of which, 126 articles related to 71 unique trials were included in the network analysis, including 34 032 patients randomly assigned to 161 treatment groups. For viral suppression at 48 weeks, compared with efavirenz, the odds ratio (OR) for viral suppression was 1.87 (95% credible interval [CrI] 1.34–2.64) with dolutegravir and 1.40 (1.02–1.96) with raltegravir; with respect to viral suppression, low-dose efavirenz was similar to all other treatments. Both low-dose efavirenz and integrase strand transfer inhibitors tended to be protective of discontinuations due to adverse events relative to normal-dose efavirenz. The most protective effect relative to efavirenz in network meta-analyses was that of dolutegravir (OR 0.26, 95% CrI 0.14–0.47), followed by low-dose efavirenz (0.39, 0.16–0.92). Owing to insufficient data, we could make no conclusions about serious adverse events. Low event rates also limited the quality of evidence with regard to mortality and AIDS defining illnesses.

Interpretation The efficacy and safety of ART has substantially improved with the introduction of newer drug classes of antiretrovirals that are now available to patients and HIV care providers. Their improved tolerance could be part of a larger solution to improve retention, which is a challenge, particularly in low-income and middle-income country settings.

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Introduction

More than 17 million people worldwide have access to antiretroviral therapy (ART) for the treatment of HIV.¹ This remarkable achievement follows decades of global efforts to scale up HIV care services after rigorous research has consistently shown efficacy of ART to reduce morbidity, mortality,² and HIV transmission.^{3,4} Further evidence now unequivocally suggests that initiation of ART earlier in disease progression improves both patient-level and population-level health outcomes.^{5,6}

WHO's consolidated guidelines on the use of antiretroviral drugs for the treatment and prevention of HIV infection offers guidance on issues related to HIV care, including the choice of treatment for ART-naïve patients. They are updated every few years to ensure the most up-to-date guidance. In the 2013 guidelines, the recommended first-line ART regimen consists of two

nucleoside or nucleotide reverse transcriptase inhibitors (NRTI) and one non-nucleoside reverse transcriptase inhibitor.⁷ The combination of efavirenz, tenofovir disoproxil fumarate, and emtricitabine or lamivudine is the preferred option.⁷

Clinical guidance on the treatment of HIV is developed through multistep processes that include safety, efficacy, equity, financial feasibility, and accessibility. WHO is not the only international agency to regularly release ART guidelines. The International Antiviral Society-USA⁸ and the US Department of Health and Human Services⁹ also provide such guidance. Although the parameters that shape recommendations differ across the agencies, such as financial considerations and the public health approach endorsed by WHO that has been necessary for the ART scale-up in low-income and middle-income countries (LMICs),

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Research in context

Evidence before this study

WHO's 2013 consolidated guidelines on the use of antiretroviral drugs for the treatment and prevention of HIV infection recommended a first-line antiretroviral therapy regimen that consists of two nucleoside or nucleotide reverse transcriptase inhibitors and one non-nucleoside reverse transcriptase inhibitor. The combination of efavirenz, tenofovir, and emtricitabine is the preferred option for first-line therapy, although a ritonavir-boosted protease inhibitor or an integrase strand transfer inhibitor-based regimen can also be used in a first-line regimen in patients with complex diseases or contraindications, or both. Evidence supporting these recommendations was based on trials published as recently as late 2012 and the evidence base was synthesised using a collection of pairwise meta-analyses. As identified by the WHO members who formulated the evidence, synthesised the research questions, and confirmed through a PubMed search for manuscripts published up to July 5, 2015 (with the search terms "HIV" AND "antiretroviral therapy" AND "randomized trial" AND ["naïve" or "first-line"]), since the publication of the

pairwise meta-analyses a few more recently published randomised controlled trials have suggested that low-dose efavirenz and dolutegravir might be preferable to standard-dose efavirenz.

Added value of this study

This study represents the first time that a network meta-analysis was used to inform a WHO guideline. Results of our analysis showed that dolutegravir and low-dose efavirenz were not only more tolerable than standard-dose efavirenz, but that they were also more effective, albeit with dolutegravir doing slightly better with respect to viral suppression efficacy and tolerability.

Implications of all available evidence

With improved efficacy and safety, and in view of WHO's public health approach to the antiretroviral therapy scale-up, steps to improve ease of care and equitability are needed before these treatments can be appointed as the preferred first-line regimens.

evidence with respect to efficacy and safety remains central to all guidelines.

Network meta-analysis is a method by which all treatment options for a disease can be assessed simultaneously. A single analysis providing an overview of a whole disease lends itself naturally to informing clinical guidelines with respect to efficacy and safety.¹⁰ By including all treatment options within a single analysis, treatments can be compared despite not having been compared in head-to-head trials. The purpose of this study was to use a network meta-analysis to assess the comparative efficacy and safety of ART regimens available at present for the treatment of HIV in ART-naïve patients.

Methods

Search strategy and selection criteria

We searched MEDLINE, Embase, and the Cochrane Central Register of Controlled Trials for randomised clinical trials published in English up to July 5, 2015, of antiretroviral regimens recommended for treatment-naïve adults and adolescents (aged 12 years or older) with HIV. The full search strategy and list of terms are listed in the appendix (p 4). The choice of age limits was determined by the question posed by WHO because they have traditionally provided recommendations for adults and adolescents. We extracted data on trial and patient characteristics (appendix p 12). Two investigators (MES and EP) did the study screening and data extraction.

Treatments were differentiated according to the specific drugs, doses, and frequency of administration. The only drugs that were regarded as interchangeable were lamivudine and emtricitabine because of their molecular

likeness and previous research.¹¹ Non-standard doses were eligible if they served as connectors (ie, were compared with two or more treatments of interest). ART regimens with a single antiviral drug and those with two drugs that included one or more NRTIs were not deemed eligible. Similarly, with the exception of boosted regimens, ART regimens with four or more drugs were not eligible (eg, a non-nucleoside reverse transcriptase inhibitor plus a protease inhibitor plus two NRTIs).

Treatments were defined according to their unique third drug, with the first two drugs being regarded as the treatment backbone (ie, two NRTIs used within the regimen). As such, the backbones needed to either be deemed balanced across treatment groups or identifiable. Trials with specific backbones within each treatment group were eligible. Trials that had mixed backbones within groups were included if either the backbones were equally distributed across groups, as shown by baseline statistics, or the backbones were selected before randomisation. Trials failing to report on backbone distribution or reporting imbalanced backbone distributions were excluded. In addition to third drugs that are still commonly used nowadays, we also included older drugs to serve as common comparators. These included drugs approved before 2000 (eg, indinavir) and newer treatments that are out of favour (eg, fosamprenavir and unboosted atazanavir). The protocol is available in the appendix (p 61).

Data analysis

We used a stepwise approach. First, we did pairwise meta-analyses with the traditional frequentist approach using the DerSimonian-Laird random-effects model,¹²

See Online for appendix

and the I^2 measure was used to gauge the degree of heterogeneity.¹³ We then did a network meta-analysis using Bayesian hierarchical models.^{14,15} All outcomes were either binary or continuous. Viral suppression and CD4 outcomes were frequently reported at multiple timepoints and were analysed separately for each of the three timepoints of interest: 24 weeks, 48 weeks, and 96 weeks. The remaining outcomes tended to be reported at a single timepoint, which varied and typically coincided with trial duration. During the feasibility assessment stage, we investigated the relation between follow-up time and outcomes (appendix pp 36–38). The odds ratios (ORs) tend to be stable over time or include an equal amount of downward and upward trends, thus supporting the mixture of varying follow-up times. We used the values at longest follow-up.

The primary outcomes were viral suppression, mortality, AIDS defining illnesses, discontinuations, discontinuations due to adverse events, and serious adverse events. Mean change in CD4 was a secondary outcome. For binary outcomes (mortality, AIDS defining illnesses, viral suppression, discontinuations, and serious adverse events), we used a logistic regression model with the logit link function and a binomial likelihood. We chose to present results as ORs for these models to avoid the ceiling effect that limits relative risks

for outcomes with proportions around 0·8 to 0·95. For continuous outcomes (eg, increase in CD4 cell count), we used linear-regression models with an identity link and normal likelihood. Estimates of comparative efficacy were represented as mean differences. Both fixed-effects and random-effects models were fit and we selected models with the deviance information criterion according to NICE conventions.¹⁶ To help identify inconsistency, we compared evidence synthesis using direct evidence with that using indirect evidence. We synthesised direct evidence using independent-means models.¹⁷ To estimate the relative treatment effects on the basis of only indirect evidence, we used edge splitting for all possible comparisons.¹⁷ Overall the deviance information criterion of the independent-means models were consistently higher than the network meta-analysis models, supporting the use of a network meta-analysis.

We chose to define the nodes in terms of the third antiretroviral drug rather than specific ART regimens. We categorised backbone regimens as tenofovir disoproxil fumarate plus lamivudine or emtricitabine (reference); abacavir plus lamivudine or emtricitabine; zidovudine plus lamivudine or emtricitabine; and other. We used group-specific meta-regression to adjust estimates according to differences in backbones according to these categories. The adjusted model served as the primary analysis; however, in outcomes for which differences in backbones were restricted to endonodal trials (non-comparative with respect to third drugs) or a few older trials with dated regimens, we used the restricted model instead. For viral suppression, the principal analysis used various thresholds, with preference for less than 50 copies per mL. Additionally, only intention-to-treat results were included in the presented findings. Finally, we used the Grading of Recommendations Assessment, Development and Evaluation (GRADE) system for rating overall quality of evidence.¹⁸ GRADE has issued guidance on network meta-analysis.¹⁹ We did all analyses using R version 3.1.2 and OpenBUGS version 3.2.3 (OpenBUGS Project Management Group; appendix p 39).

Role of the funding source

The funder of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report. The corresponding author had full access to all the data in the study and had final responsibility for the decision to submit for publication.

Results

We identified a total of 5865 citations through database searches and other sources; of these, 513 were selected for full-text review (appendix pp 6–11). Ultimately 126 manuscripts were included in the analysis pertaining to 71 unique trials (figure 1).^{20–152} Overall, trials were of generally good quality with low risk of bias (appendix pp 18–19). None of the studies included in our analysis were restricted to adolescents only.

For the R statistics program see
<http://www.r-project.org/>

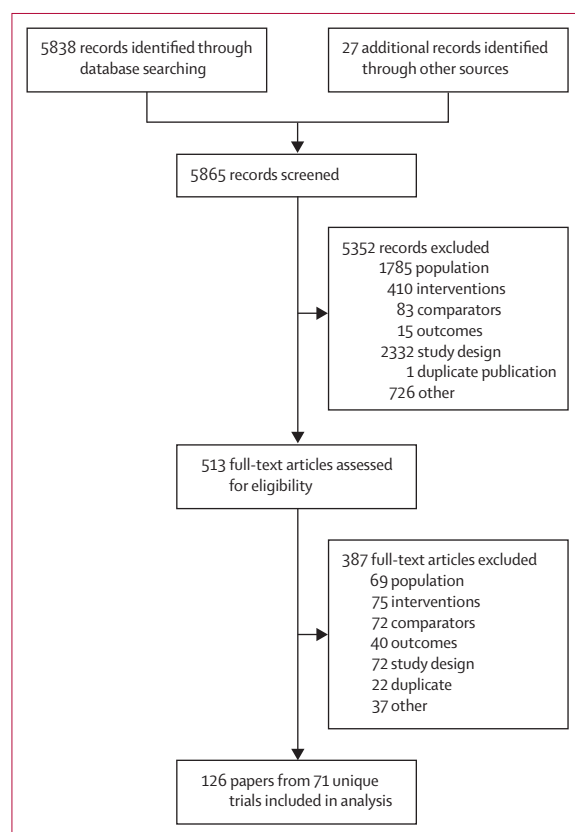


Figure 1: Study selection

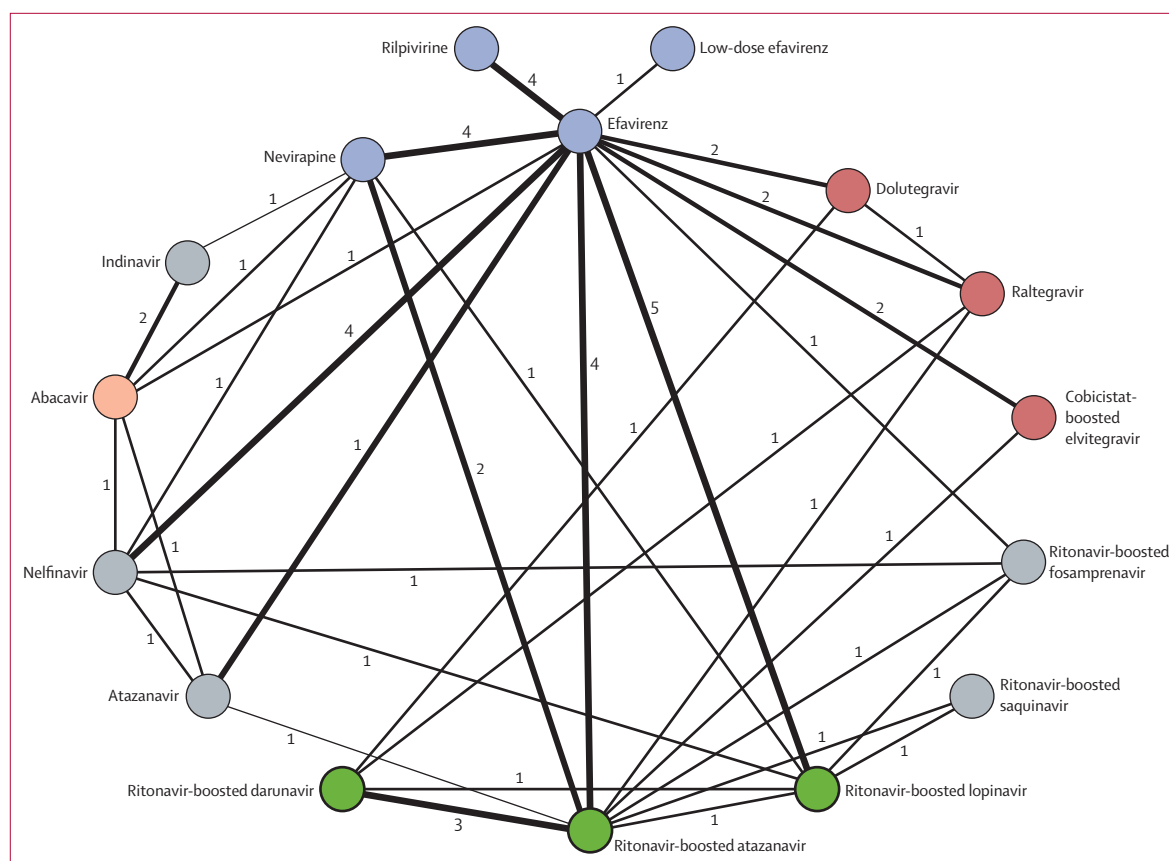


Figure 2: Network of eligible comparisons between treatments

Overall, the network included 34 032 patients randomly assigned to 161 treatment groups across 71 trials. Circles (nodes) in the diagrams represent individual treatments, lines between circles represent availability of head-to-head evidence between two treatments, and the numbers on the lines are the number of randomised clinical trials informing each head-to-head comparison. The colours represent antiretroviral classes, with grey representing protease inhibitors that are no longer commonly used in practice, which are included in the network as connectors.

The network included 34 032 patients randomly assigned to 161 treatment groups. The resulting overall network was well connected (figure 2). Efavirenz was the most well connected treatment, with both integrase strand transfer inhibitors (INSTI) and ritonavir-boosted protease inhibitor drugs directly connected to it. Several trials compared INSTI and ritonavir-boosted protease inhibitor drugs head-to-head, and several sources of indirect evidence were available to inform comparisons between efavirenz, INSTI drugs, and ritonavir-boosted protease inhibitor drugs. Low-dose efavirenz was only linked to efavirenz head-to-head via one trial and was therefore compared with all other treatments indirectly.

A total of 70 trials including 31 404 patients and comprising 154 treatment groups, pertaining to 16 third drugs, were included for the assessment of viral suppression at 24, 48, and 96 weeks. 30 trials reported viral suppression at 24 weeks (appendix p 20), 64 at 48 weeks, and 25 at 96 weeks (figure 3). Dolutegravir was significantly better than efavirenz at 48 weeks and at 96 weeks. Raltegravir was the only other treatment statistically superior to efavirenz at these timepoints.

Lopinavir fared worst and was inferior to normal-dose efavirenz in addition to all INSTI drugs. INSTI drugs, particularly dolutegravir, showed better viral suppression outcomes against most other third drugs; however, a comparison between INSTI drugs found that cobicistat-boosted elvitegravir was inferior to dolutegravir and that this comparison was statistically significant at 96 weeks. Low-dose efavirenz was statistically superior to nevirapine only at 96 weeks. In this analysis, rilpivirine was also superior to nevirapine.

A total of 47 trials including 22 634 patients and 476 deaths were included in the mortality analysis. However, most of the deaths were reported in studies published before 2000 that are of little interest to this analysis. Among the comparisons of interest, only 28 deaths were reported, making a network meta-analysis unreliable (appendix p 26).

With respect to AIDS defining illnesses, the evidence base was quite sparse. Although cobicistat-boosted elvitegravir had statistically higher odds of AIDS defining illnesses than most treatments, this calculation was based on only five events (appendix p 27).

EFV	1.90 (1.40-2.59)	1.45 (1.07-1.95)	1.10 (0.77-1.59)	0.69 (0.48-1.03)	0.93 (0.74-1.18)	0.99 (0.71-1.40)	0.49 (0.30-0.82)	1.19 (0.73-1.95)	1.34 (0.92-1.94)
1.87 (1.34-2.64)	DTG	0.76 (0.56-1.03)	0.58 (0.37-0.92)	0.36 (0.24-0.56)	0.49 (0.35-0.69)	0.52 (0.37-0.74)	0.26 (0.14-0.47)	0.63 (0.35-1.11)	0.71 (0.44-1.14)
1.40 (1.02-2.59)	0.75 (0.53-1.05)	RAL	0.76 (0.49-1.18)	0.48 (0.32-0.73)	0.64 (0.47-0.88)	0.68 (0.48-0.97)	0.34 (0.19-0.61)	0.82 (0.46-1.44)	0.93 (0.57-1.49)
1.28 (0.87-1.89)	0.68 (0.41-1.14)	0.91 (0.56-1.50)	EVG/c	0.63 (0.39-1.03)	0.84 (0.59-1.22)	0.90 (0.57-1.44)	0.45 (0.24-0.83)	1.09 (0.58-1.98)	1.22 (0.72-2.03)
0.76 (0.59-0.98)	0.40 (0.27-0.60)	0.54 (0.37-0.78)	0.59 (0.38-0.92)	LPV/r	1.34 (0.96-1.85)	1.43 (1.00-2.00)	0.72 (0.39-1.27)	1.73 (0.91-3.11)	1.94 (1.12-3.21)
0.90 (0.74-1.10)	0.48 (0.33-0.69)	0.64 (0.46-0.89)	0.70 (0.48-1.04)	1.18 (0.92-1.54)	ATV/r	1.07 (0.78-1.48)	0.54 (0.31-0.92)	1.28 (0.73-2.20)	1.45 (0.92-2.22)
0.91 (0.66-1.28)	0.49 (0.33-0.72)	0.65 (0.45-0.94)	0.71 (0.44-1.16)	1.21 (0.87-1.69)	1.02 (0.74-1.40)	1.02 (0.74-1.40)	0.50 (0.27-0.90)	1.20 (0.65-2.17)	1.36 (0.80-2.21)
0.87 (0.70-1.07)	0.46 (0.32-0.68)	0.62 (0.43-0.89)	0.68 (0.44-1.04)	1.15 (0.85-1.54)	0.97 (0.76-1.23)	0.95 (0.65-1.37)	NVP	2.42 (1.18-4.88)	2.72 (1.46-5.11)
1.16 (0.67-2.02)	0.62 (0.33-1.17)	0.82 (0.44-1.55)	0.90 (0.46-1.77)	1.52 (0.83-2.59)	1.29 (0.72-2.31)	1.26 (0.67-2.39)	1.33 (0.74-2.40)	low EFV	1.13 (0.61-2.13)
1.18 (0.90-1.55)	0.63 (0.41-0.98)	0.85 (0.55-1.28)	0.92 (0.57-1.48)	1.57 (1.07-2.25)	1.32 (0.93-1.83)	1.29 (0.83-1.98)	1.36 (0.96-1.92)	1.02 (0.56-1.87)	RPV

■ Treatment ■ 48 week network results, OR (95% CI) □ 96 week network results, OR (95% CI)

Figure 3: Random-effects network meta-analyses of the relative efficacy of antiretrovirals for viral suppression

Data are OR (95% CI) of the row treatment relative to the column treatment (eg, the effect of dolutegravir relative to efavirenz is 1.87 with respect to viral suppression at 48 weeks). Bold values indicate comparisons that are statistically significant. ORs above 1 indicate higher efficacy in viral suppression. OR=odds ratio. EFV=efavirenz. DTG=dolutegravir. RAL=raltegravir. EVG/c=cobicistat-boosted elvitegravir. LPV/r=ritonavir-boosted lopinavir. ATV/r=ritonavir-boosted atazanavir. DRV/r=ritonavir-boosted darunavir. NVP=nevirapine. RPV=rilpivirine.

EFV	0.84 (0.49-1.43)	0.98 (0.67-1.45)	2.04 (0.29-54.74)	1.02 (0.82-1.27)	0.90 (0.65-1.26)	0.58 (0.34-0.98)	1.87 (1.31-2.69)	1.00 (0.55-1.85)	0.84 (0.61-1.14)
0.26 (0.14-0.47)	DTG	1.17 (0.77-1.77)	2.43 (0.31-66.55)	1.22 (0.69-2.17)	1.08 (0.57-2.02)	0.69 (0.33-1.46)	2.23 (1.17-4.25)	1.20 (0.53-2.69)	1.00 (0.54-1.86)
0.46 (0.24-0.86)	1.74 (0.84-3.60)	RAL	2.08 (0.28-56.73)	1.04 (0.67-1.63)	0.92 (0.55-1.54)	0.59 (0.31-1.14)	1.91 (1.33-2.75)	1.02 (0.50-2.11)	0.85 (0.52-1.41)
0.70 (0.41-1.16)	2.65 (1.23-5.71)	1.52 (0.67-3.55)	EVG/c	0.50 (0.02-3.62)	0.44 (0.02-3.05)	0.28 (0.01-2.18)	0.91 (0.03-6.77)	0.49 (0.02-3.89)	0.41 (0.02-3.02)
1.35 (0.87-2.10)	5.16 (2.65-10.12)	2.97 (1.38-6.42)	1.94 (1.03-3.70)	LPV/r	0.88 (0.60-1.30)	0.57 (0.34-0.93)	1.83 (1.21-2.78)	0.98 (0.51-1.88)	0.82 (0.56-1.20)
0.89 (0.60-1.33)	3.40 (1.75-6.77)	1.95 (0.94-4.18)	1.28 (0.77-2.24)	0.66 (0.42-1.06)	ATV/r	0.64 (0.35-1.16)	2.07 (1.30-3.30)	1.11 (0.55-2.24)	0.93 (0.59-1.46)
0.47 (0.24-0.88)	1.79 (0.87-3.60)	1.02 (0.43-2.44)	0.67 (0.31-1.47)	0.35 (0.19-0.62)	0.53 (0.27-0.99)	DRV/r	3.22 (1.71-6.10)	1.72 (0.77-3.87)	1.44 (0.78-2.66)
1.58 (0.96-2.61)	6.00 (2.89-12.70)	3.47 (1.53-7.80)	2.26 (1.18-4.44)	1.17 (0.68-2.03)	1.76 (1.12-2.77)	3.36 (1.64-7.03)	NVP	0.54 (0.26-1.09)	0.45 (0.28-0.72)
0.39 (0.16-0.92)	1.49 (0.52-4.20)	0.85 (0.20-2.52)	0.56 (0.20-1.51)	0.29 (0.11-0.76)	0.44 (0.16-1.12)	0.83 (0.41-1.90)	0.25 (0.09-0.65)	low EFV	0.83 (0.42-1.66)
0.41 (0.26-0.63)	1.57 (0.76-3.25)	0.90 (0.42-1.96)	0.59 (0.30-1.18)	0.31 (0.16-0.56)	0.46 (0.25-0.82)	0.88 (0.41-1.90)	0.26 (0.13-0.50)	1.06 (0.40-2.81)	RPV

■ Treatment ■ Discontinuations because of adverse events, OR (95% CrI)
□ Treatment emergent serious adverse events, OR (95% CrI)

Figure 4: Network meta-analyses comparing antiretrovirals in terms of discontinuation due to adverse events (random effects) and treatment emergent serious adverse events (fixed effects)

Data are OR (95% CrI) of the row treatment relative to the column treatment (eg, the effect of dolutegravir relative to efavirenz is 0.26 with respect to discontinuation due to adverse events). Bold values indicate comparisons that are statistically significant. ORs above 1 indicate higher risk of discontinuation due to adverse event. OR=odds ratio. CrI=credible interval. EFV=efavirenz. DTG=dolutegravir. RAL=raltegravir. EVG/c=cobicistat-boosted elvitegravir. LPV/r=ritonavir-boosted lopinavir. ATV/r=ritonavir-boosted atazanavir. DRV/r=ritonavir-boosted darunavir. NVP=nevirapine. RPV=rilpivirine.

A total of 14 trials including 5033 patients and comprising 27 treatment groups formed the evidence network for treatment-related serious adverse events (appendix p 28). With half of the trials including at least one group with no events and a very sparse network, meta-analyses and network meta-analyses were of little use. Nothing substantive could be determined from the data. In view of the sparse data on treatment-related serious adverse events, we analysed drug-emergent serious adverse events, which included 39 trials and 20650 patients (figure 4). Most treatments were not statistically different from one another, with the exception

of nevirapine, which was worse than all other treatments but one. Cobicistat-boosted elvitegravir had the largest effect size relative to all other treatments.

We also did a random-effects network meta-analysis for discontinuations due to adverse events (figure 4). Low-dose efavirenz, ritonavir-boosted darunavir, and INSTIs tended to be protective of discontinuations due to adverse events relative to standard-dose efavirenz. The most protective effect relative to efavirenz was that of dolutegravir, followed by low-dose efavirenz. Low-dose efavirenz was not statistically differentiable from dolutegravir, but the estimated effect suggested higher proportions of discontinuation due to adverse events. When using all-cause discontinuations (appendix p 29), we found that all treatment effects relative to efavirenz were attenuated, with only dolutegravir and raltegravir remaining significantly better than efavirenz.

A total of 66 trials including 28728 patients were included in the assessment of CD4 cell counts at 24, 48, and 96 weeks. 31 trials reported CD4 cell counts at 24 weeks, 61 at 48 weeks, and 30 at 96 weeks (appendix pp 22-24). In the 24 week analysis, only one comparison reached statistical significance, with dolutegravir showing a mean increase in CD4 count of 34 cells per μL (95% credible interval [CrI] 3.46-64.76) compared with standard-dose efavirenz. At 48 weeks, a larger and better connected network found that all three INSTI drugs (dolutegravir, raltegravir, and cobicistat-boosted elvitegravir) were superior to standard-dose efavirenz, with the mean differences in CD4 counts of 22.93 cells per μL for dolutegravir, 20.09 cells per μL for raltegravir, and 18.45 cells per μL for cobicistat-boosted elvitegravir. We also found that both dolutegravir and raltegravir were superior to ritonavir-boosted protease inhibitors atazanavir and darunavir. The 48 week analysis also found that low-dose efavirenz was superior to standard-dose efavirenz (mean difference 25.49; 95% CrI 6.66-44.37).

Direct and indirect evidence showed high agreement throughout the analyses, thus meeting the condition of consistency. The exceptions were two networks, viral suppression at 24 weeks and discontinuations, which each contained a loop presenting evidence of inconsistency. In both cases, this was resolved by removing the unboosted-atazanavir trials. Finally, for most outcomes, the random-effects model was chosen indicating some degree of heterogeneity, as would be expected.

Discussion

Our systematic review and network meta-analysis, conducted to inform a component of the new WHO Consolidated HIV Treatment Guidelines, found that although efavirenz plus two NRTIs as a backbone remains a safe and efficacious regimen, other treatments are in some regards comparatively superior. The evidence suggests that dolutegravir is superior to standard-dose

efavirenz, both as third drug, with respect to viral suppression and rates of discontinuation, whereas low-dose efavirenz is superior to standard-dose efavirenz with respect to rates of discontinuation and gains in CD4 cell counts. A research question posed by WHO in anticipation of the guideline development was how INSTIs compared with efavirenz, and to this end our results suggest a clear hierarchy within the INSTI class, with dolutegravir being the most efficacious, followed by raltegravir, and then elvitegravir. Several reasons remain outside of efficacy and safety for standard-dose efavirenz to continue being the favoured first-line drug worldwide; however, the reported benefits of dolutegravir and low-dose efavirenz signal the possibility for future change.

There are several implications and considerations related to these findings. Our study suggests that some alternative first-line treatments are superior to the present WHO recommendation of efavirenz plus an NRTI backbone; however, complexity of care must also be considered. The present recommended regimen is available in a fixed-dose combination, with easy once a day dosing. Dolutegravir is available in a fixed-dose combination that includes emtricitabine and abacavir, requiring screening for HLA-B*5701 to predict hypersensitivity reaction.¹⁵³ The screening adds a layer of complexity to care and increases the burden on the patient. Because of the complexity of screening for abacavir, the fixed-dose formulation for dolutegravir in LMICs will probably include tenofovir disoproxil fumarate plus a lamivudine or emtricitabine backbone. Although this combination was present in the evidence base, it was uncommon. Nonetheless, our regression adjustments suggest improved efficacy with the combination, implying a possible optimum combination with dolutegravir (ie, tenofovir disoproxil fumarate plus a lamivudine or emtricitabine backbone plus dolutegravir). Despite the improved efficacy and safety, issues exist regarding the feasibility of scaling up a first-line regimen containing dolutegravir. Primarily, WHO's public health approach to the ART scale-up would imply switching the ART regimen for millions of patients, which is likely to come with many logistic and clinical problems. Nonetheless, the present ART scale-up remains imperfect because mortality continues to be high in LMICs.¹⁵⁴ The improved tolerability of dolutegravir, efavirenz 400 mg, and rilpivirine could be part of a solution to improve retention to HIV care and ultimately reduce mortality, but as HIV care has taught us, it is combinations of interventions that are needed.

Our study found that efavirenz 400 mg could also be an important candidate for a first-line ART regimen. However, evidence for this finding is restricted to ENCORE1²² and issues regarding its performance in patients co-infected with tuberculosis and women who are pregnant or breastfeeding have not been studied in clinical trials. Evidence from pharmacokinetic studies show that this absence of research in these populations

might not be an issue; nonetheless, although the public health approach to ART scale-up calls for limited treatment options, adaptation of dosing for particular populations might be necessary.

Our study has several strengths. First and foremost, the use of a network meta-analysis provides advantages over the methods used in past reviews to inform ART guidelines. These methods allow for the simultaneous analysis of all treatments eligible for consideration as the preferred first-line regimen, thus simplifying the overall evidence synthesis process. Second, the combination of direct and indirect evidence improves estimation by reducing the uncertainty bounds about estimates that had strong agreement between direct and indirect comparisons. Some effect sizes were not deemed significant in pairwise meta-analysis, but were significant in network meta-analysis. Third, our group-specific meta-regression allowed us to gain understanding of the effect of backbone regimens and, in turn, to combine data on efficacy despite differences in backbones. This use of group-specific meta-regression was particularly relevant to this research question given that the SINGLE trial,¹⁴⁸ a large phase 3 trial providing direct evidence to the question at hand, would have been excluded had we restricted inclusion criteria to trials only comparing third drugs.

Our study also has limitations. First, some important outcomes were limited by a low number of events, particularly mortality and AIDS defining illnesses, affecting the precision of our estimates with respect to these outcomes. Second, the evidence base was limited to short follow-up times, mostly up to 2 years. ART is taken over a lifetime, so it would be of great interest to understand how these treatments compare after 10 years or more. Does superiority in the short term necessarily translate to long-term advantages? We note that for the purpose of safety, long-term results are more readily available for populations that are not restricted to ART-naïve patients and that this was also investigated for the purpose of guideline development. Third, treatment-related adverse events were both inconsistently defined and inconsistently reported. For example, some trials reported only grade 2–4 treatment-related adverse events, whereas others defined serious adverse events as grade 3 or higher and some trials only reported adverse events if they led to a discontinuation of the study drug. One way we resolved this discrepancy was by analysing all drug-emergent adverse events. Finally, the CrIs for the CD4 cell count outcome were at times large, despite convergence of the Markov Chain Monte Carlo chains, providing little insight into comparative effectiveness.

In conclusion, our systematic literature review found that among ART-naïve patients, the use of an INSTI plus two NRTIs, particularly dolutegravir and raltegravir, has superior efficacy and tolerance to regimens of efavirenz plus two NRTIs, and that low-dose efavirenz is non-inferior to standard-dose efavirenz. Their improved

tolerance could be part of a larger solution to improve retention, which is a challenge, particularly in LMIC settings.

Contributors

SK and EJM had full access to all of the data in the study. SK takes responsibility for the integrity of the data, the accuracy of the data analysis, and the final decision to submit for publication. SK, KT, EJM, MV, MD, MES, and NF contributed to the study concept and design. SK, KT, JIF, and EP contributed to data acquisition, analysis, and interpretation. SK, KT, EJM, MES, and JIF drafted the manuscript. SK, KT, EJM, MV, MD, MES, NF, JIF, EP, NB, and SN critically revised the manuscript for important intellectual content. SK, KT, EJM, and NB did the statistical analysis. EJM obtained funding. SK, MV, NF, and EJM contributed to administrative, technical, or material support. EJM, MD, MV, NF, and NB supervised the study.

Declaration of interests

We declare no competing interests.

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